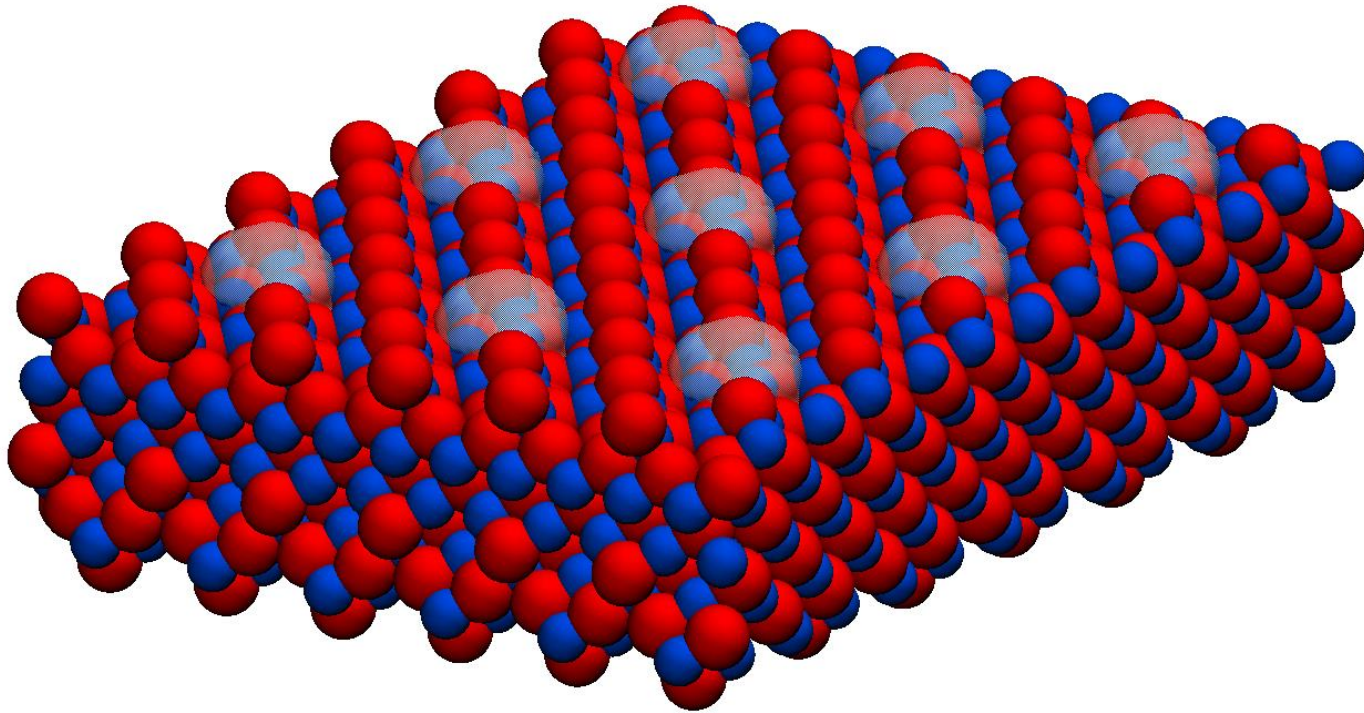


Reminders about planewaves and Brillouin zone sampling

Pseudopotentials, PAW and other basis sets



Sergey Levchenko

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Moscow, Russia*

Reminder: electronic-structure problem

$$\hat{h}\psi_i = \varepsilon_i\psi_i$$

$$\hat{h} = -\frac{1}{2}\nabla^2 + V_{ext}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' + \hat{V}_{XC}$$

ψ_i - unknown functions (one-electron wave functions)

$$\psi_i(\mathbf{r}) = \sum_p C_{ip} \phi_p(\mathbf{r}) \quad \text{- basis set expansion}$$

$\phi_p(\mathbf{r})$ - known functions (basis functions)

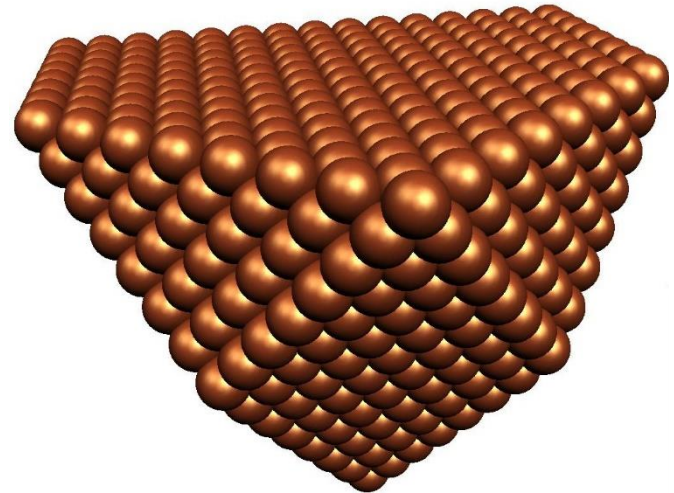
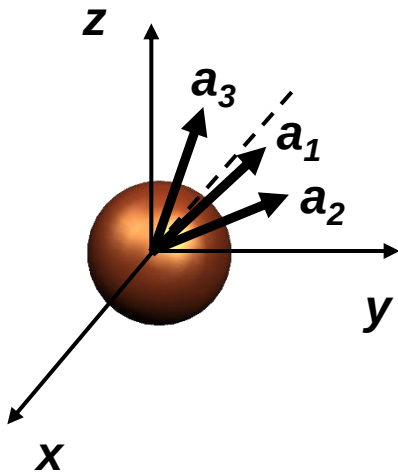


Generalized eigenvalue problem:

$$\sum_q \langle \phi_p | \hat{h}[\mathbf{C}] | \phi_q \rangle C_{iq} = \varepsilon_i \sum_q \langle \phi_p | \phi_q \rangle C_{iq}$$

Extended (periodic) systems

There are 10^{20} electrons per 1 mm^3 of bulk Cu



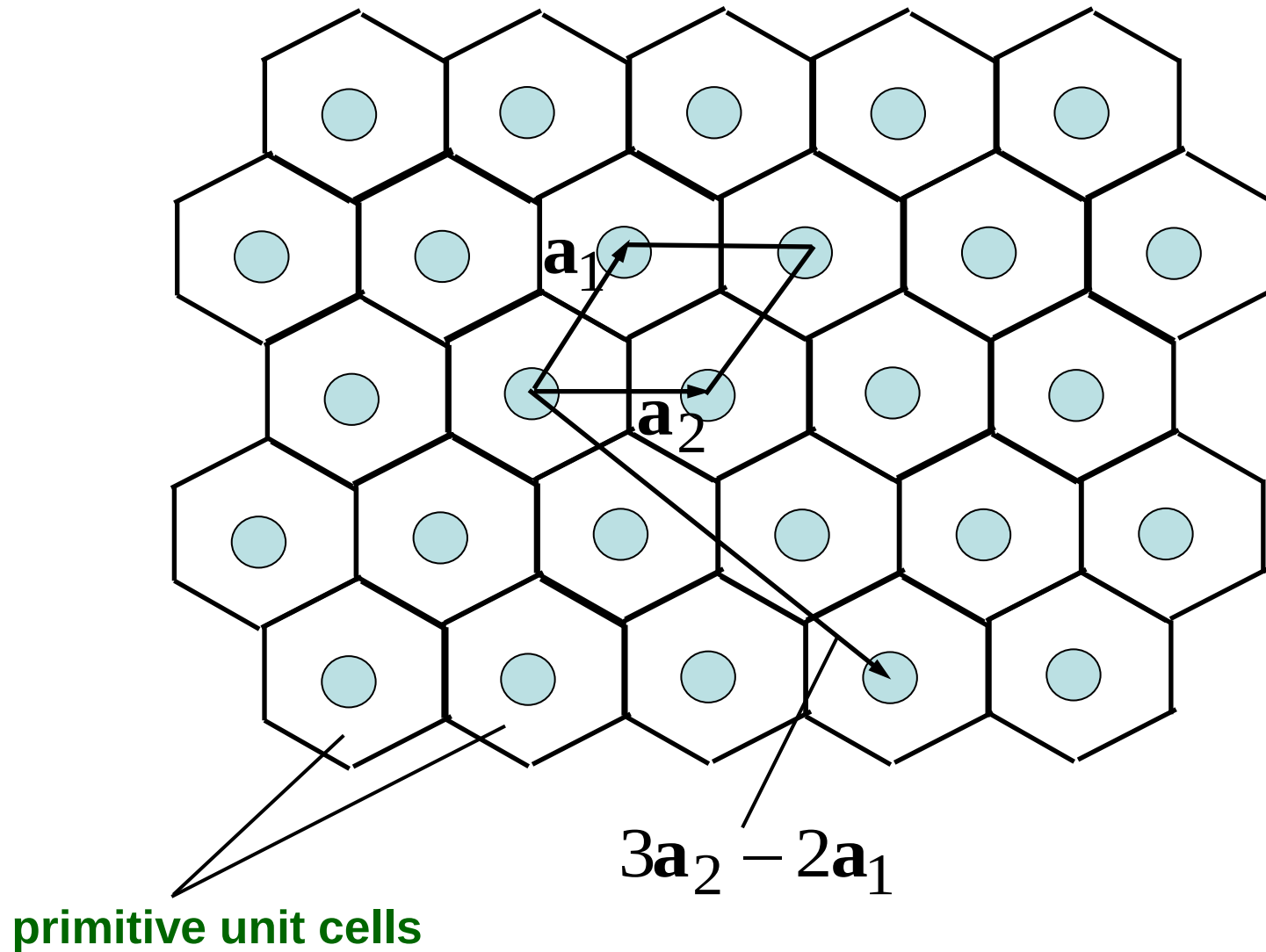
Position of every atom in the crystal (Bravais lattice):

$$\mathbf{r}(n_1, n_2, n_3) = \mathbf{r}(0,0,0) + n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$$

lattice vector: $\mathbf{R}(n_1, n_2, n_3) = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$

$$n_1, n_2, n_3 = 0, \pm 1, \pm 2, \dots$$

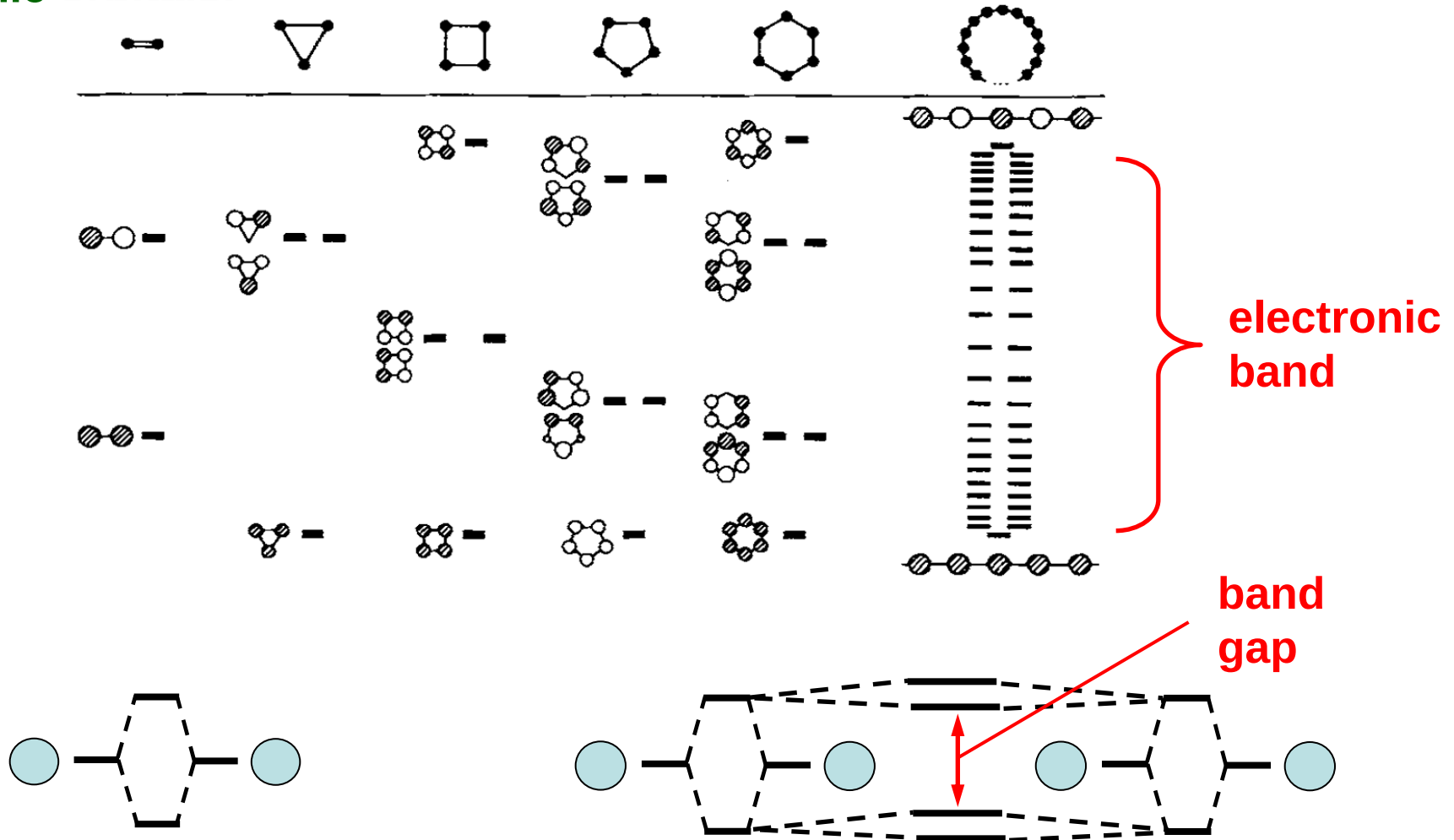
Example: two-dimensional Bravais lattice



The shape of the primitive unit cell is not unique

From molecules to solids

Electronic bands as limit of bonding and anti-bonding combinations of atomic orbitals:



Bloch's theorem

Periodic potential $U(\mathbf{r} + \mathbf{R}) = U(\mathbf{r})$
(translational symmetry)

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

In an infinite periodic solid, the solutions of the one-particle Schrödinger equations must behave like

$$\psi(\mathbf{r} + \mathbf{R}) = \exp(i\mathbf{k}\mathbf{R})\psi(\mathbf{r})$$

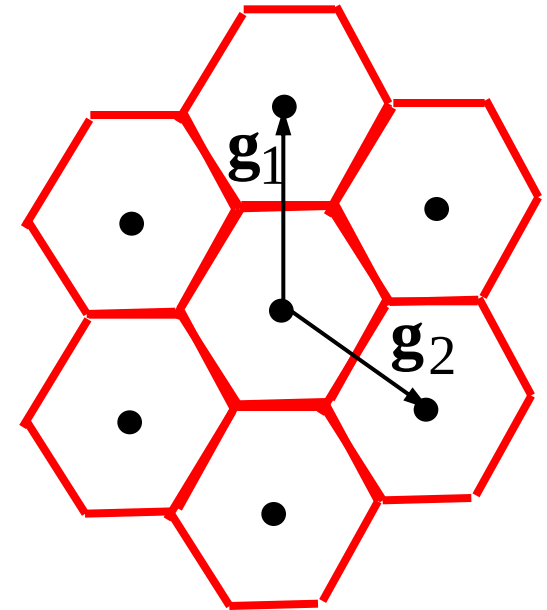
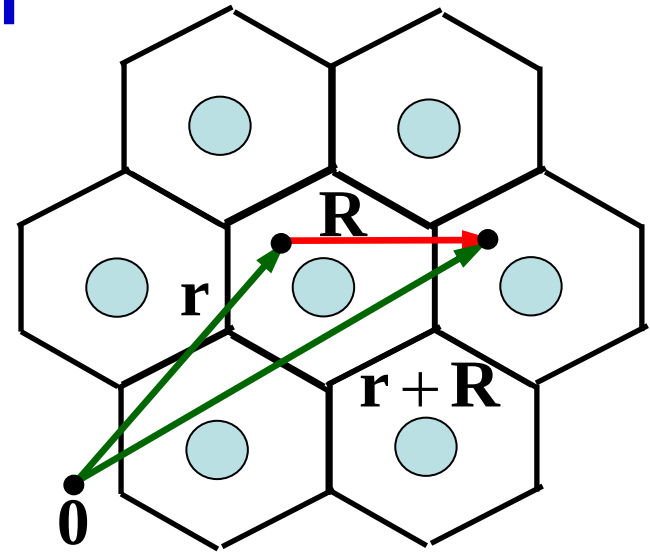
Index $\mathbf{k}$ is a vector in *reciprocal space*

$$\mathbf{k} = x_1 \mathbf{g}_1 + x_2 \mathbf{g}_2 + x_3 \mathbf{g}_3 \quad \mathbf{g}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij}$$

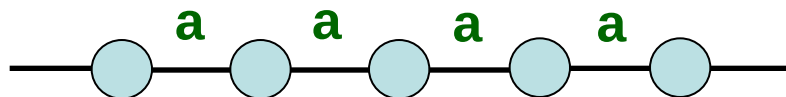
$$\mathbf{g}_l = 2\pi \frac{\mathbf{a}_m \times \mathbf{a}_n}{\Omega} \quad \text{-- reciprocal lattice vectors}$$

Consequently:

$$\psi(\mathbf{r}) = \exp(i\mathbf{k}\mathbf{r})u(\mathbf{r}), \quad u(\mathbf{r} + \mathbf{R}) = u(\mathbf{r})$$



The meaning of k

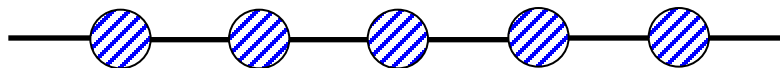


chain of hydrogen atoms

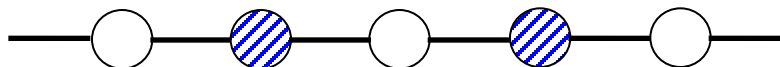
$$\psi_k = \sum_j \exp(ikx_j) \chi_{1s}(j \cdot a)$$

k shows the phase with which the orbitals are combined:

$k = 0:$ $\psi_0 = \sum_j \exp(0) \chi_{1s}(j \cdot a) = \chi_{1s}(a) + \chi_{1s}(2a) + \dots$



$k = \frac{\pi}{a}:$ $\psi_0 = \sum_j \exp(i\pi \cdot j) \chi_{1s}(j \cdot a) = -\chi_{1s}(a) + \chi_{1s}(2a) - \chi_{1s}(3a) + \dots$



k is a symmetry label and a node counter, and also represents electron momentum

Bloch's theorem: consequences

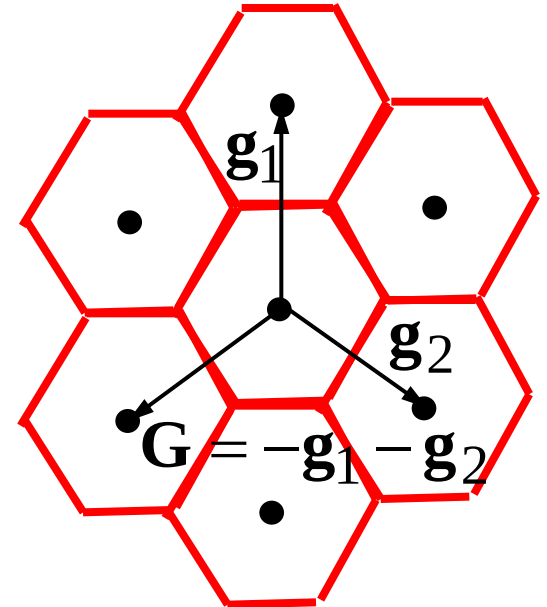
In a periodic system, the solutions of the Schrödinger equations are characterized by an integer number n (called *band index*) and a vector $\mathbf{k}$:

$$\hat{h} \psi_{n\mathbf{k}} = \varepsilon_{n\mathbf{k}} \psi_{n\mathbf{k}}$$

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \exp(i\mathbf{k}\mathbf{r}) u_{n\mathbf{k}}(\mathbf{r}), \quad u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})$$

For any reciprocal lattice vector

$$\mathbf{G} = n_1 \mathbf{g}_1 + n_2 \mathbf{g}_2 + n_3 \mathbf{g}_3$$



$$\psi_{n\mathbf{k}+\mathbf{G}} = \exp(i\mathbf{k}\mathbf{r}) \underbrace{[u_{\mathbf{k}+\mathbf{G}} \exp(i\mathbf{G}\mathbf{r})]}_{\text{a lattice-periodic function } \tilde{u}} = \exp(i\mathbf{k}\mathbf{r}) \tilde{u} = \psi_{n'\mathbf{k}}$$

a Bloch state
at $\mathbf{k}+\mathbf{G}$ with
index n

a lattice-periodic
function $\tilde{u}$

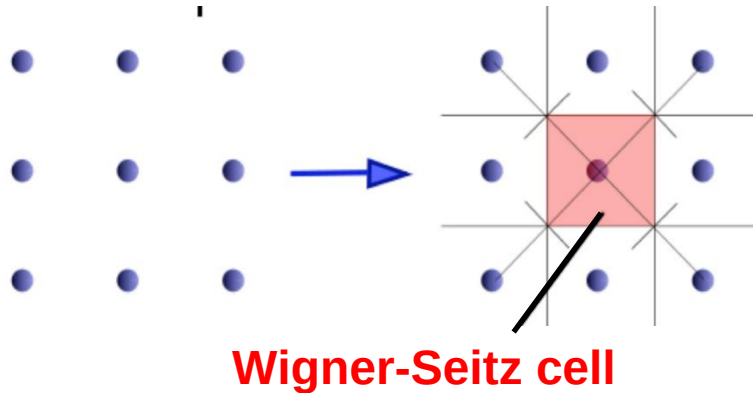
a Bloch state at $\mathbf{k}$ with
a different index n'

Can choose to consider only $\mathbf{k}$ within single primitive
unit cell in reciprocal space

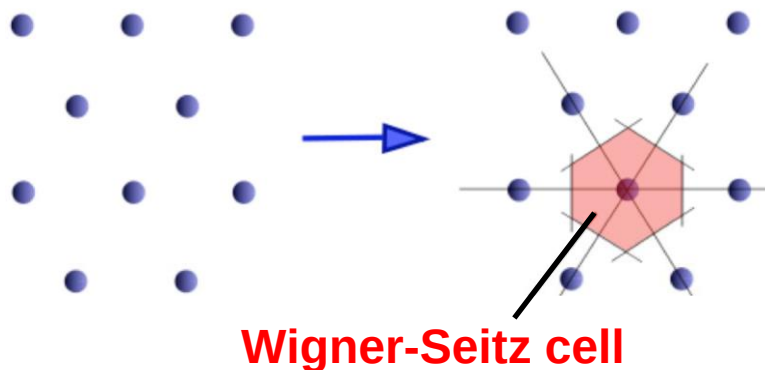
Brillouin zones

A conventional choice for the reciprocal lattice unit cell

For a square lattice

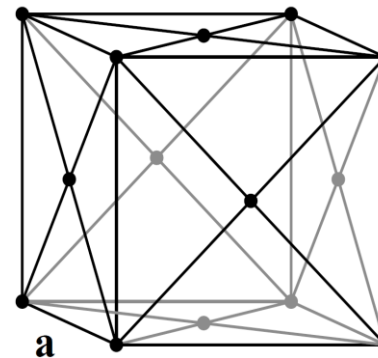


For a hexagonal lattice

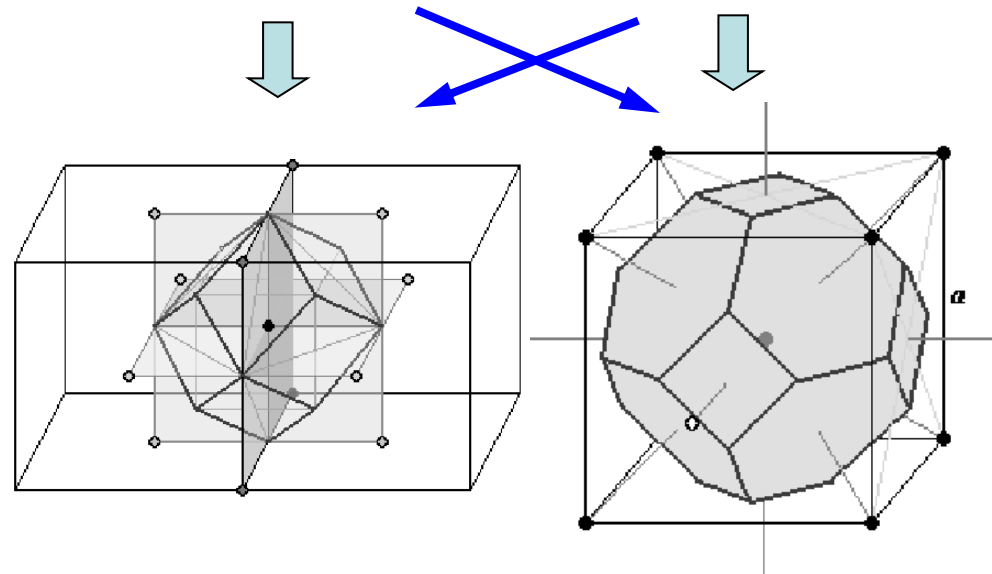
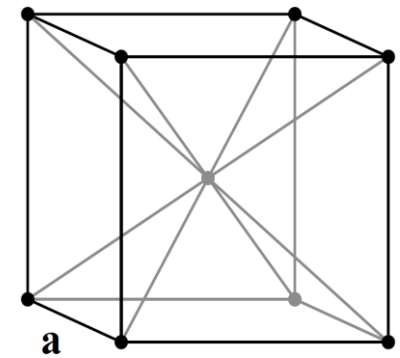


In three dimensions:

Face-centered
cubic (fcc) lattice



Body-centered
cubic (bcc) lattice



Time-reversal symmetry

For Hermitian $\hat{h}$, $\varepsilon_{n\mathbf{k}}$ can be chosen to be *real*

$$\hat{h}\psi_{n\mathbf{k}}(\mathbf{r}) = \varepsilon_{n\mathbf{k}}\psi_{n\mathbf{k}}(\mathbf{r}) \Rightarrow \hat{h}\psi_{n\mathbf{k}}^*(\mathbf{r}) = \varepsilon_{n\mathbf{k}}\psi_{n\mathbf{k}}^*(\mathbf{r})$$

From Bloch's theorem:

$$\psi_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = \exp(i\mathbf{k}\mathbf{R})\psi_{n\mathbf{k}}(\mathbf{r}) \Rightarrow \psi_{n\mathbf{k}}^*(\mathbf{r} + \mathbf{R}) = \exp(-i\mathbf{k}\mathbf{R})\psi_{n\mathbf{k}}^*(\mathbf{r})$$



$$\psi_{n\mathbf{k}}^* = \psi_{n(-\mathbf{k})} \quad \varepsilon_{n(-\mathbf{k})} = \varepsilon_{n\mathbf{k}}$$

**Electronic states at $\mathbf{k}$ and $-\mathbf{k}$ are at least doubly degenerate
(in the absence of magnetic field)**

Basis sets

$$\psi_i(\mathbf{r}) = \sum_p C_{ip} \varphi_p(\mathbf{r})$$

Commonly used basis sets:

plane waves $\exp(i\mathbf{k} \cdot \mathbf{r})$ **(delocalized, analytic integrals)**

gaussians $x^i y^j z^k \exp(-\alpha r^2)$ **(localized, analytic integrals)**

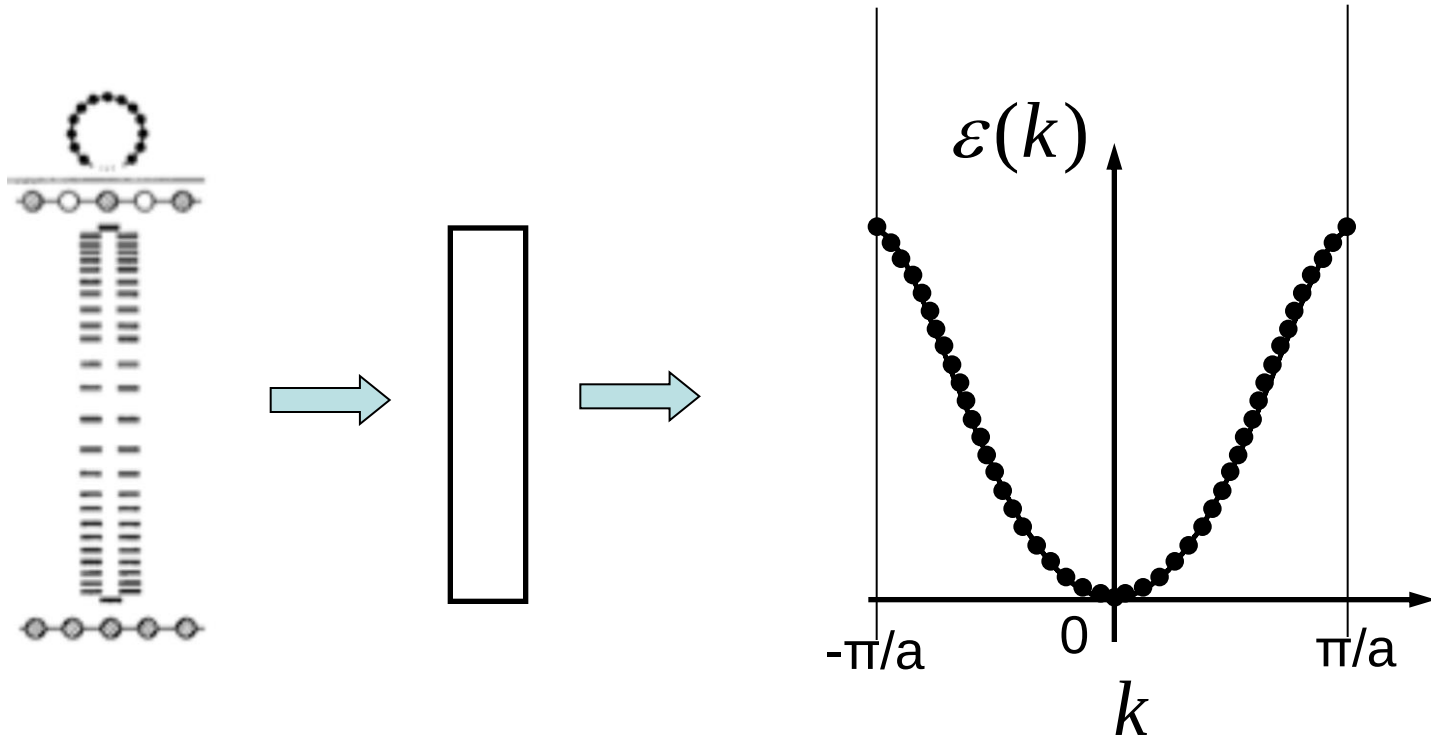
Slater-type $x^i y^j z^k \exp(-\alpha r)$ **(localized, nuclear cusp)**

Numeric atomic orbitals $\frac{u(r)}{r} Y_{lm}(\theta, \phi)$ **(localized, flexible)**

grid-based $\delta(\mathbf{r} - \mathbf{r}_i)$ **(localized)**

Core electrons are often treated separately
(pseudopotentials, plane-wave + localized basis)

Electronic band structure

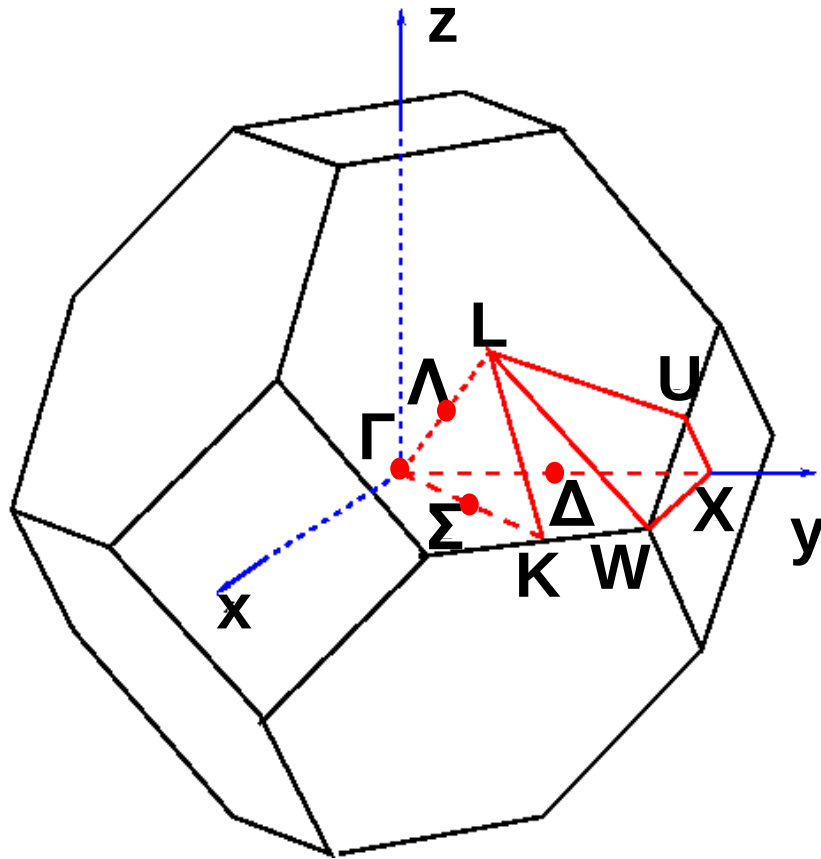


Band structure represents dependence of $\varepsilon_{n\mathbf{k}}$ on $\mathbf{k}$

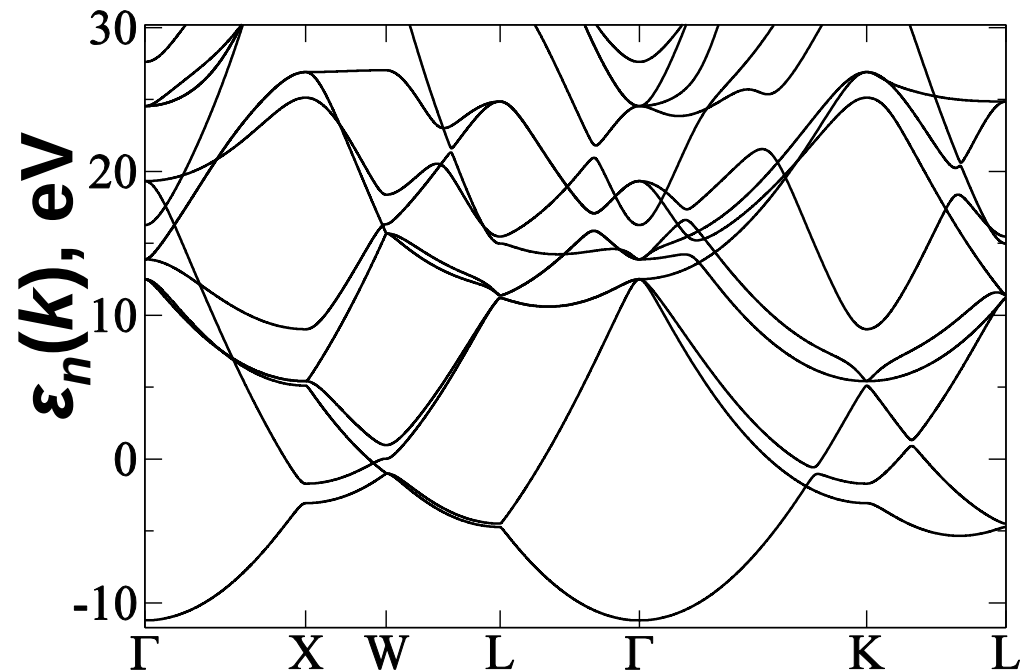
For a periodic (infinite) crystal, there is an infinite number of states for each band index n , differing by the value of $\mathbf{k}$

Electronic band structure in three dimensions

Brillouin zone of the fcc lattice



Al band structure (DFT-PBE)



By convention, $\varepsilon_{n\mathbf{k}}$ are measured (angular-resolved photoemission spectroscopy, ARPES) and calculated along lines in k-space connecting points of high symmetry

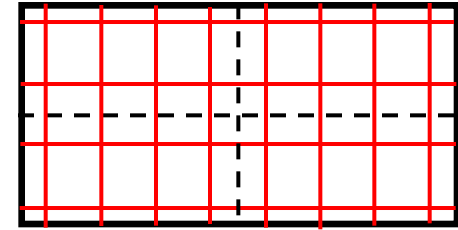
Finite k-point mesh

Charge densities and other quantities are represented by Brillouin zone integrals:

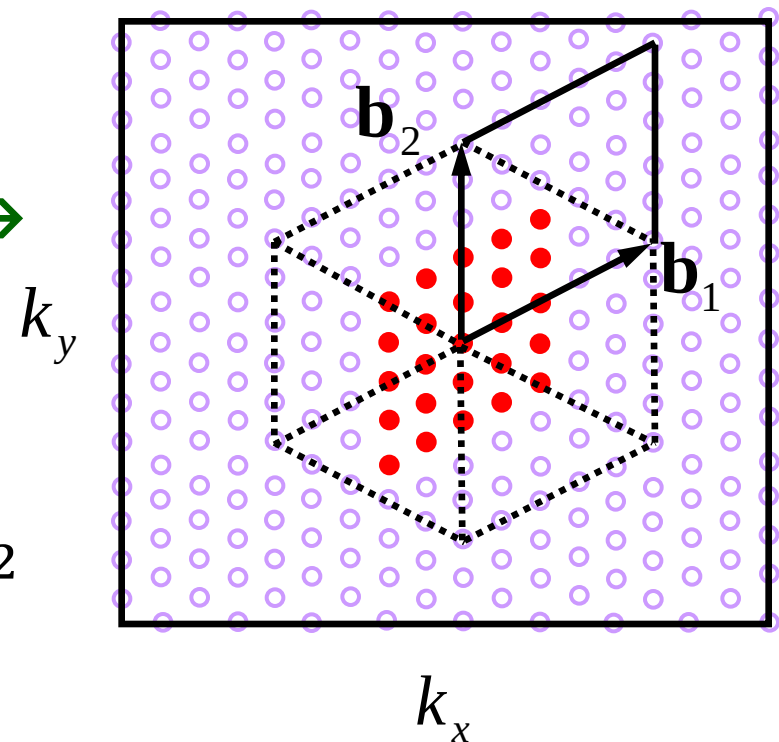
$$n(\mathbf{r}) = \sum_{j=0}^{\infty} \int_{\Omega_{BZ}} f_{jk} |\psi_{jk}(\mathbf{r})|^2 \frac{d^3k}{\Omega_{BZ}}$$

$\psi_{n\mathbf{k}}$, $\varepsilon_{n\mathbf{k}}$ – *smooth* functions of $\mathbf{k} \rightarrow$ can use a **finite** mesh, and then *interpolate* and/or use perturbation theory to calculate integrals

$$n(\mathbf{r}) \approx \sum_{j=0}^{\infty} \sum_{m=1}^{N_{kpt}} w_m |\psi_{jk_m}(\mathbf{r})|^2$$



8x4 Monkhorst-Pack grid



Finite k-point mesh: symmetry

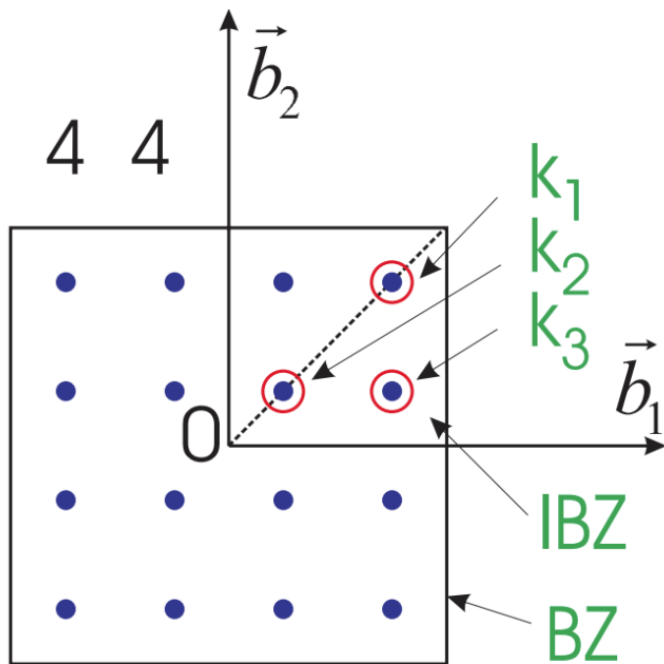
$$\mathbf{k}_{prs} = u_p \mathbf{g}_1 + u_r \mathbf{g}_2 + u_s \mathbf{g}_3$$

$$u_r = \frac{2r - q_r - 1}{2q_r}, r = 1, 2, \dots, q_r$$

reciprocal lattice vectors

number of k-point in direction r

Example: a quadratic 2D lattice $q_1 = q_2 = 4$, i.e., 16 k-points in total



Only 3 symmetry non-equivalent k-points

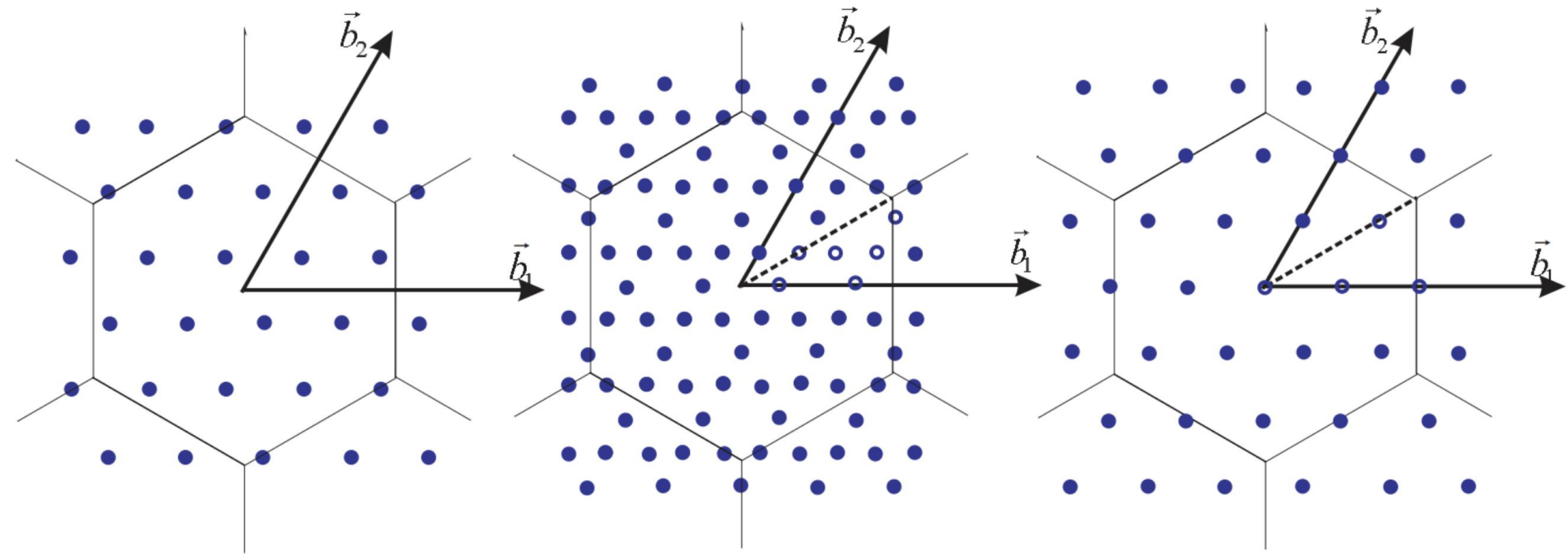
$$4 \times k_1 = \left(\frac{3}{8}, \frac{3}{8} \right) \rightarrow w_1 = \frac{4}{16}$$

$$4 \times k_2 = \left(\frac{1}{8}, \frac{1}{8} \right) \rightarrow w_2 = \frac{4}{16}$$

$$8 \times k_3 = \left(\frac{3}{8}, \frac{1}{8} \right) \rightarrow w_3 = \frac{8}{16}$$

$$\int_{\Omega_{BZ}} F(\mathbf{k}) d^2 k \approx \frac{1}{4} F(\mathbf{k}_1) + \frac{1}{4} F(\mathbf{k}_2) + \frac{1}{2} F(\mathbf{k}_3)$$

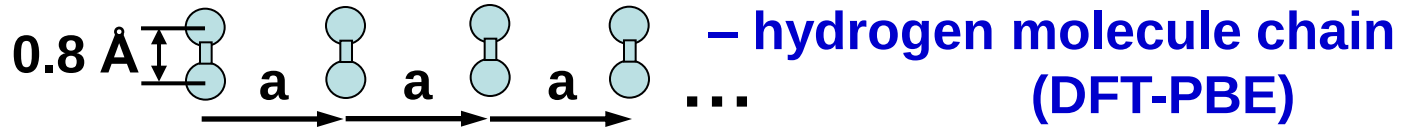
Finite k-point mesh: symmetry



In certain cell geometries (e.g., hexagonal cells) even shifted meshes break the symmetry - symmetrization results in non-uniform distribution of k-points

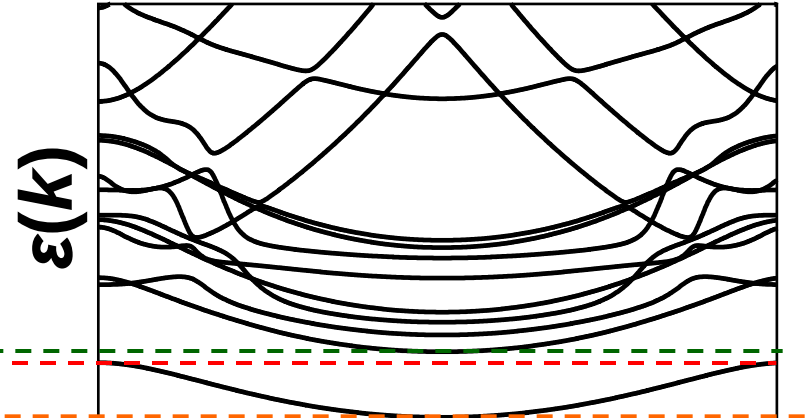
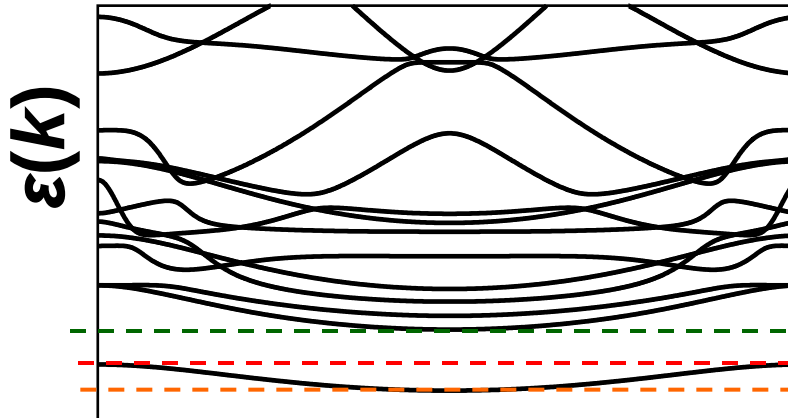
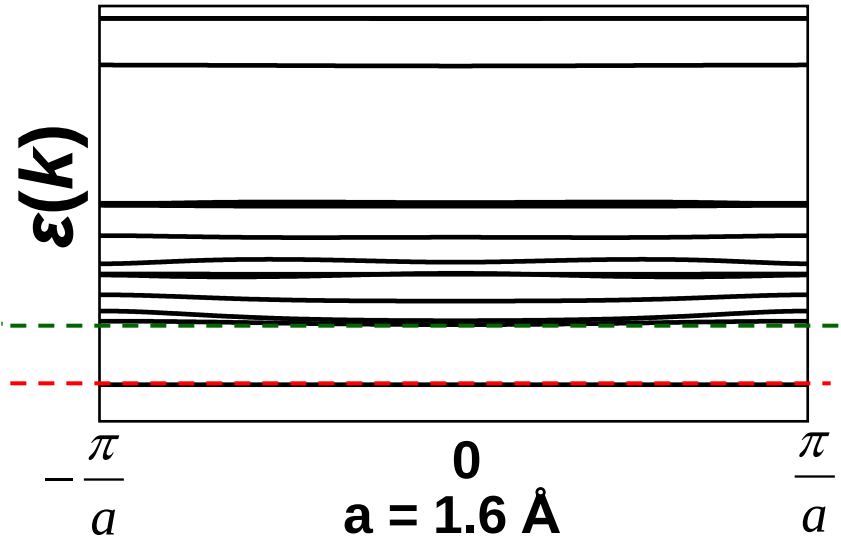
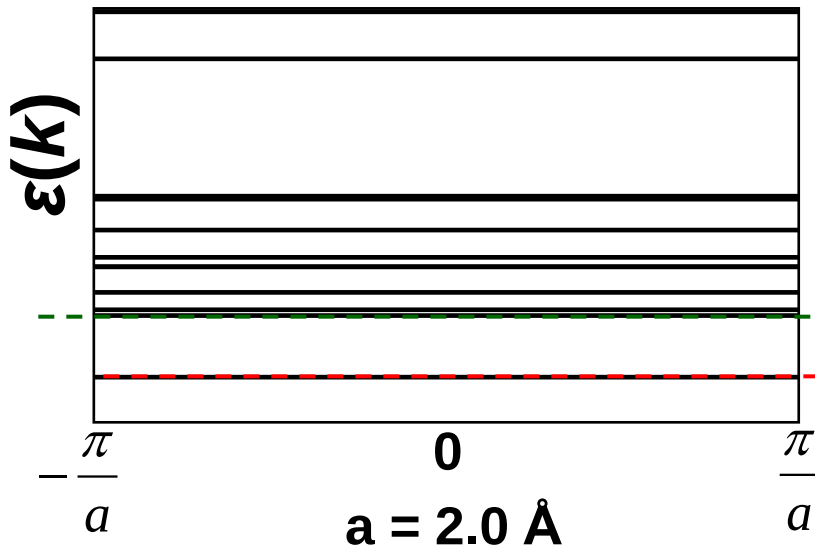
Γ -point centered meshes preserve the symmetry

Band gap and band width (dispersion)



$a = 10.0 \text{ Å}$

$a = 5.0 \text{ Å}$

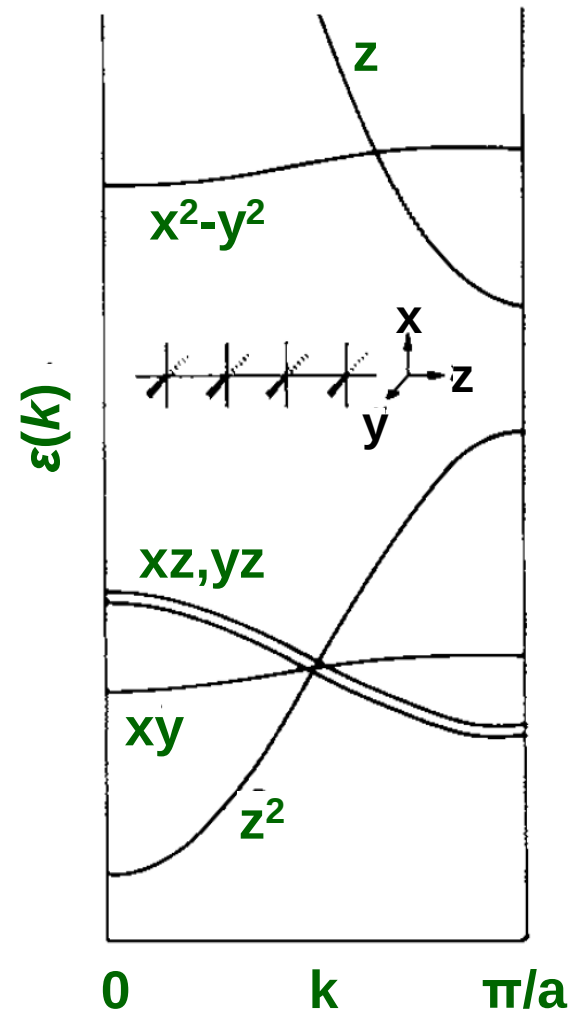
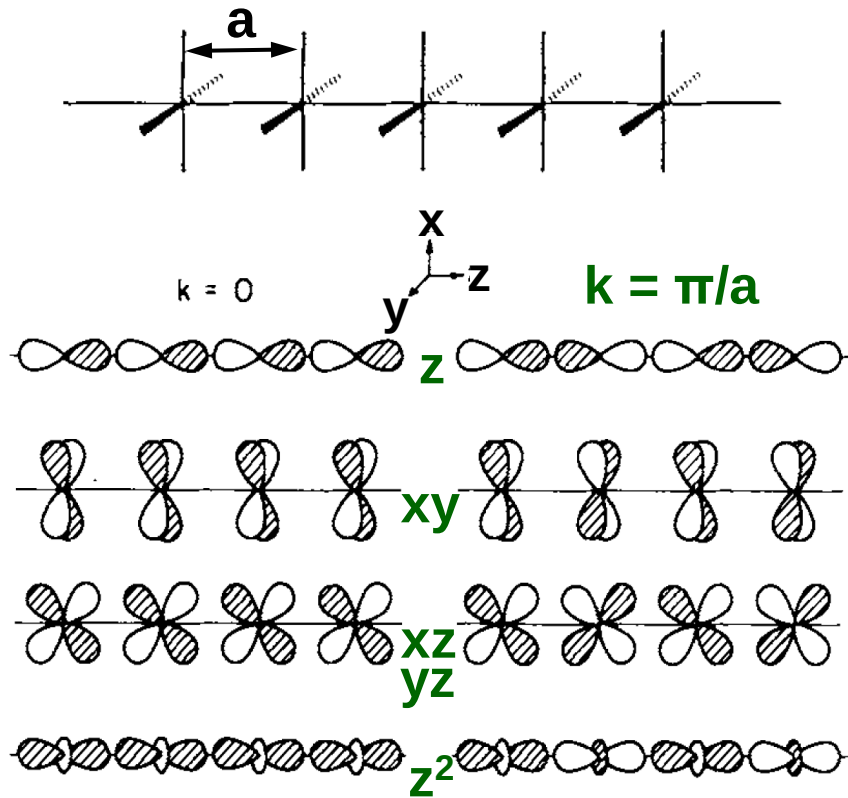


Overlap between interacting orbitals determines band gap and band width

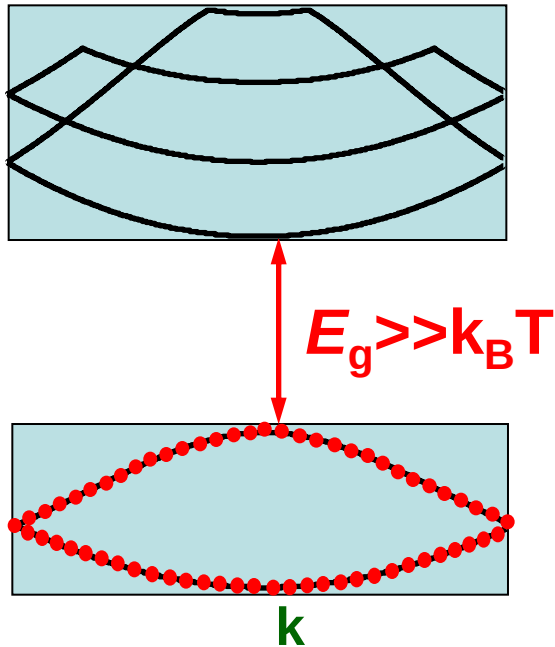
Band structure – test example

Orbital energies are smooth functions of k

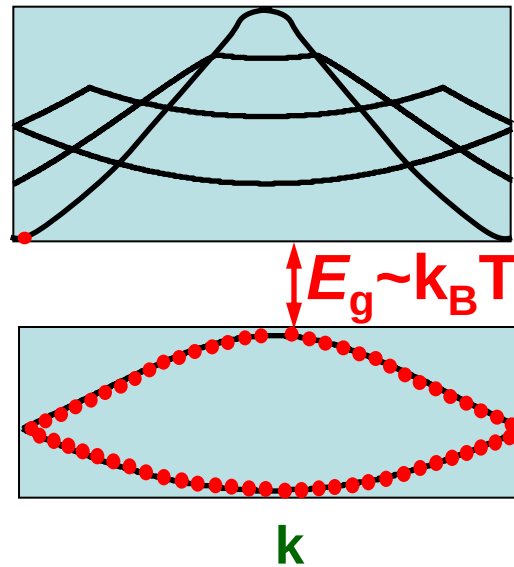
Example: chain of Pt-L₄ complexes (K₂[Pt(CN)₄])



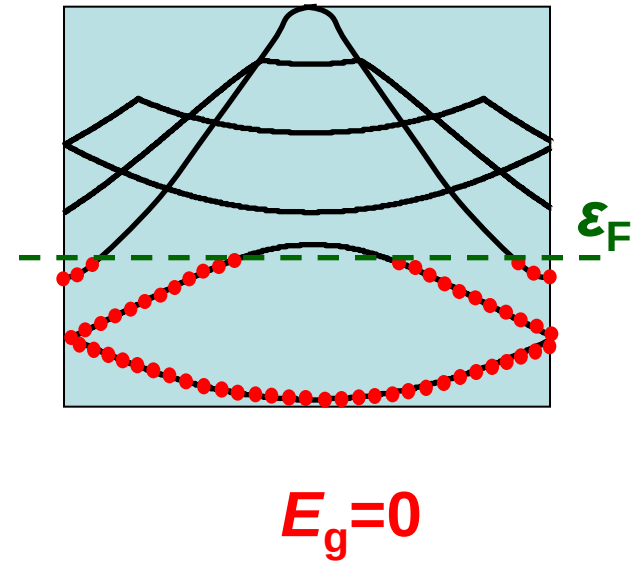
Insulators, semiconductors, and metals



Insulators (MgO, NaCl, ZnO,...)



Semiconductors (Si, Ge,...)



Metals (Cu, Al, Fe,...)

In a metal, some (at least one) energy bands are only partially occupied

The Fermi energy ϵ_F separates the highest occupied states from lowest unoccupied

Fermi surface

Plotting the relation

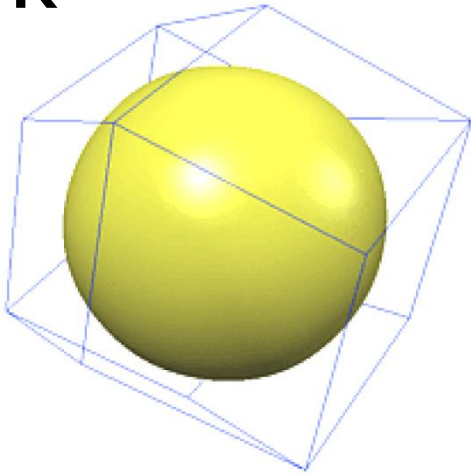
$$\varepsilon_n(\mathbf{k}) = \varepsilon_F$$

in reciprocal space for different n yields different parts of the Fermi surface

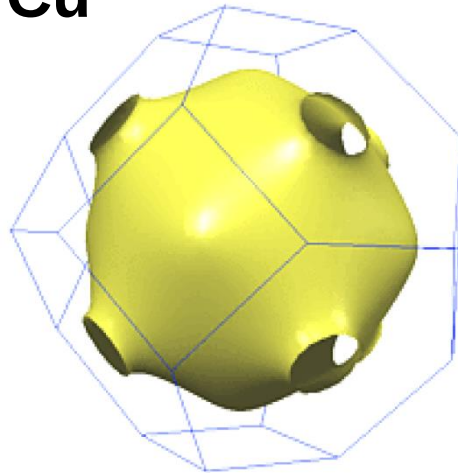
For free electrons, Fermi surface is a sphere

$$\frac{\hbar^2 \mathbf{k}^2}{2m_e} = \varepsilon_F$$

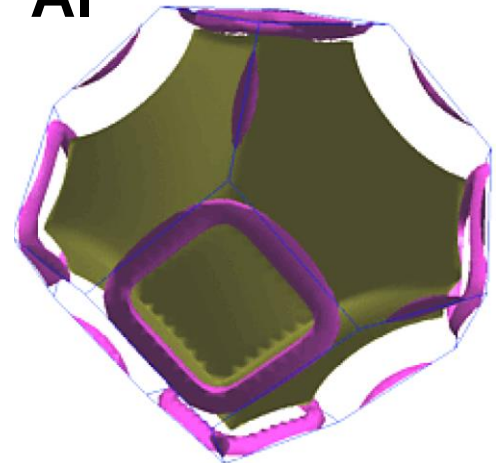
K



Cu



Al



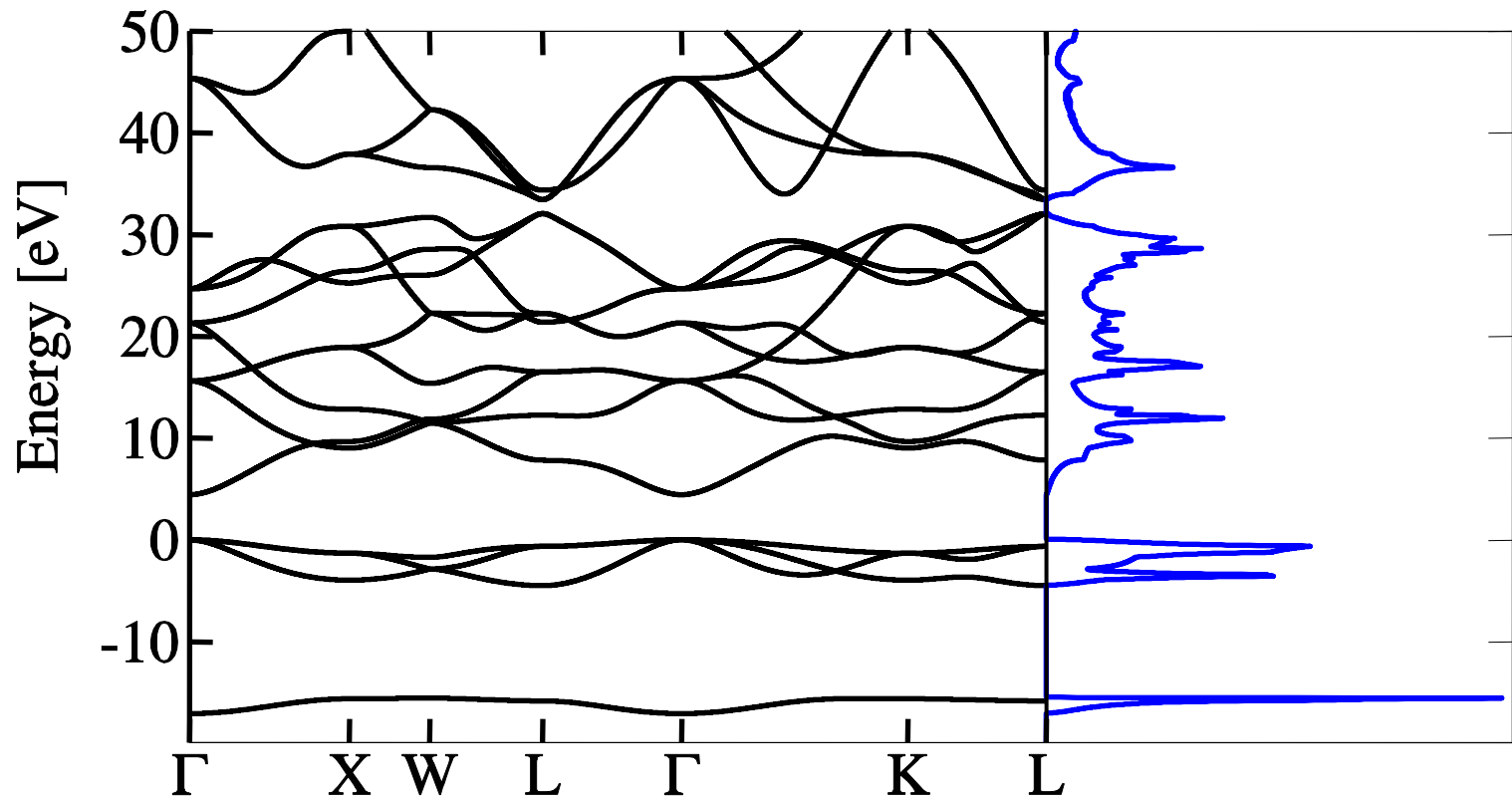
Periodic table of Fermi surfaces: <http://www.phys.ufl.edu/fermisurface/>

The grid used in k -space must be sufficiently fine to accurately sample the Fermi surface

Density Of States (DOS)

Number of states in energy interval $d\varepsilon$ per unit volume, $\frac{1}{\Omega d\varepsilon} \sum_{\varepsilon}^{\varepsilon+d\varepsilon} 1$

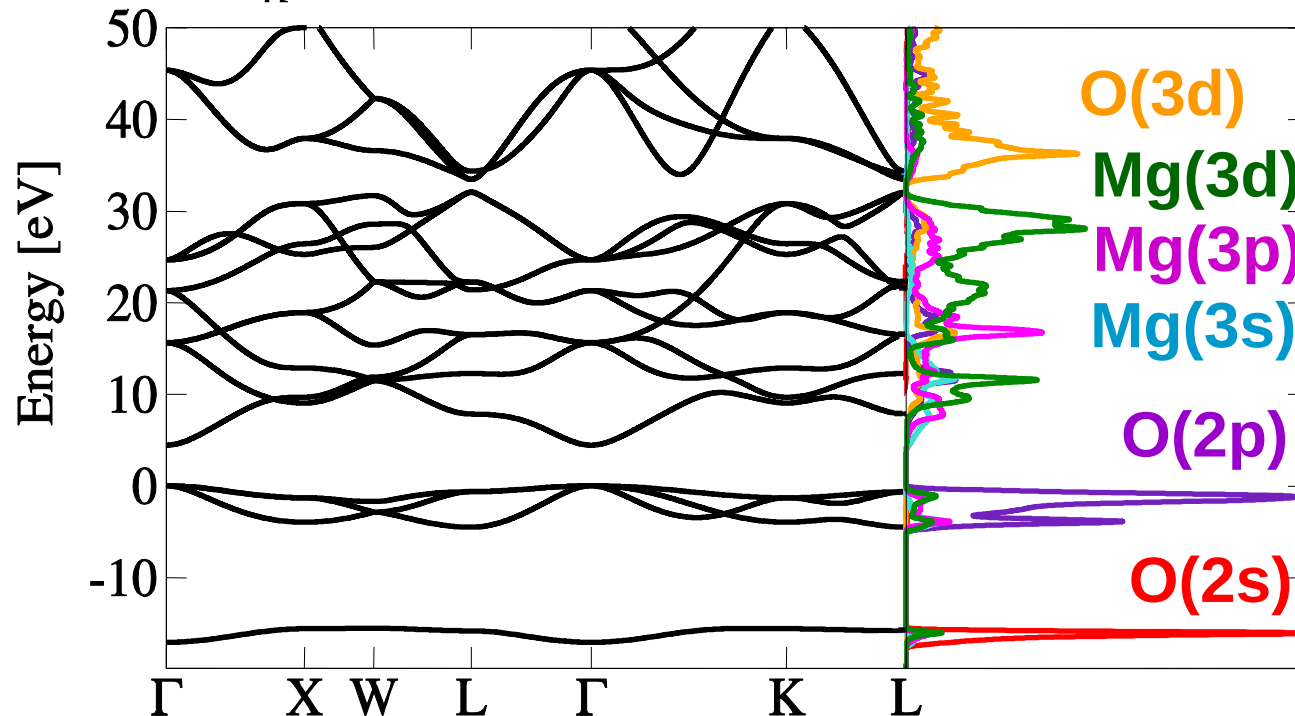
$$g(\varepsilon) = \sum_n \int_{\Omega_{\text{BZ}}} \delta(\varepsilon - \varepsilon_n(\mathbf{k})) d^3k \approx \sum_n \sum_{j=1}^{N_{\text{kpt}}} w_j \delta(\varepsilon - \varepsilon_n(\mathbf{k}_j))$$



Atom-Projected Density Of States (APDOS)

Decomposition of DOS into contributions from different atomic functions φ_i :

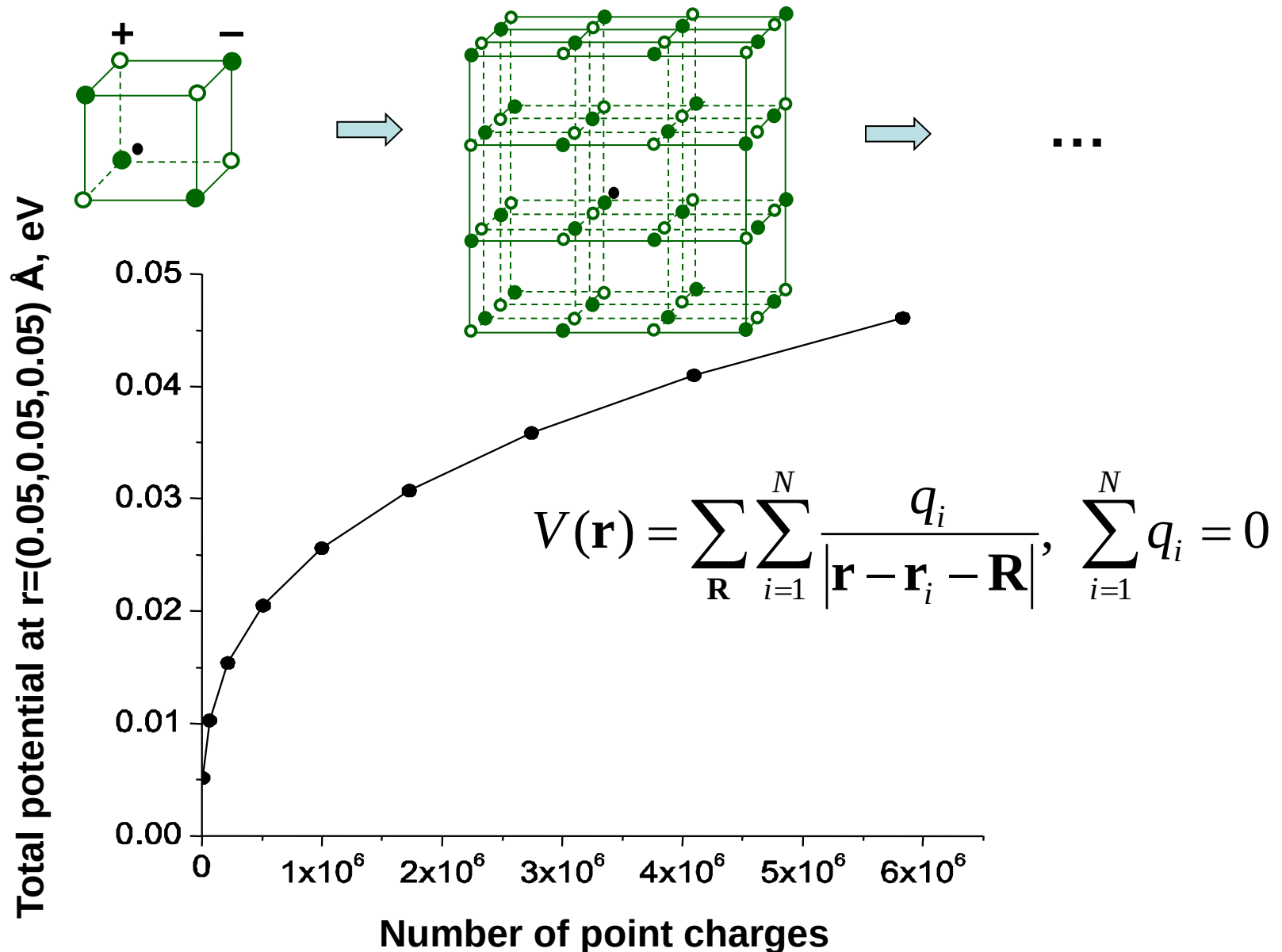
$$g_i(\varepsilon) = \sum_n \int_{\Omega_{\text{BZ}}} \left| \int \varphi_i(\mathbf{r}) \psi_{n\mathbf{k}}(\mathbf{r}) d^3r \right|^2 \delta(\varepsilon - \varepsilon_n(\mathbf{k})) d^3k$$



Recovery of the chemical interpretation in terms of orbitals

Qualitative analysis tool; ambiguities must be resolved by truncating the r-integral or by Löwdin orthogonalization of φ_i

Potential of an array of point charges



Convergence of the potential with number of charges is extremely slow

Ewald summation

$$V(\mathbf{r}) = \sum_{\mathbf{R}} \sum_{i=1}^N \frac{q_i}{|\mathbf{r} - \mathbf{r}_i - \mathbf{R}|} + \frac{1}{\eta\sqrt{\pi}} \exp\left(-(\mathbf{r} - \mathbf{r}_i)^2 / \eta^2\right)$$



$$\nabla^2 V(\mathbf{r}) = -4\pi\rho(\mathbf{r})$$

(Poisson's equation)

$$V_1(\mathbf{r}) = \sum_{i,\mathbf{R}} q_i \left(\frac{\text{erfc}(|\mathbf{r} - \mathbf{r}_i - \mathbf{R}| / \eta)}{|\mathbf{r} - \mathbf{r}_i - \mathbf{R}|} \right)$$

Decays fast with $|\mathbf{R}|$

$$V_2(\mathbf{r}) = \sum_{i,\mathbf{G} \neq \mathbf{0}} q_i \frac{4\pi}{\mathbf{G}^2 \Omega} \exp\left(-\frac{\eta^2 \mathbf{G}^2}{4} + i\mathbf{G} \cdot (\mathbf{r} - \mathbf{r}_i)\right)$$

Decays fast with $|\mathbf{G}|$

Diverges at $\mathbf{G} = \mathbf{0}$, but divergence

is cancelled for $\sum_i q_i = 0$

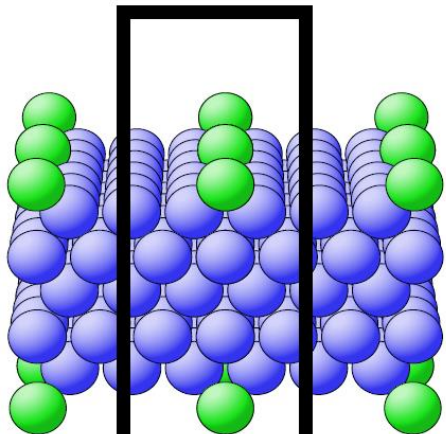
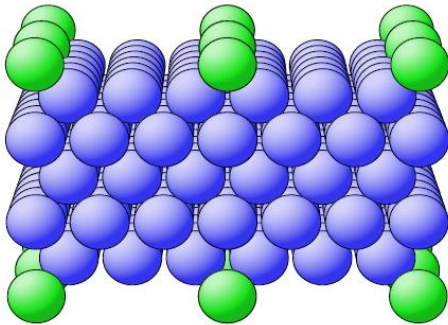
There is no universal potential energy reference (like vacuum level) for 3D periodic systems – important when comparing different systems

The supercell approach

Can we benefit from periodic modeling of non-periodic systems?

Yes, for interfaces (surfaces) and wires (also with adsorbates), and defects (especially for concentration or coverage dependences)

Supercell approach to surfaces
(slab model)



supercell

- Approach accounts for the lateral periodicity
- Sufficiently broad vacuum region to decouple the slabs
- Sufficient slab thickness to mimic semi-infinite crystal
- Semiconductors: saturate dangling bonds on the back surface
- Non-equivalent surfaces: use dipole correction
- Alternative: cluster models (for defects and adsorbates)

Plane wave basis set

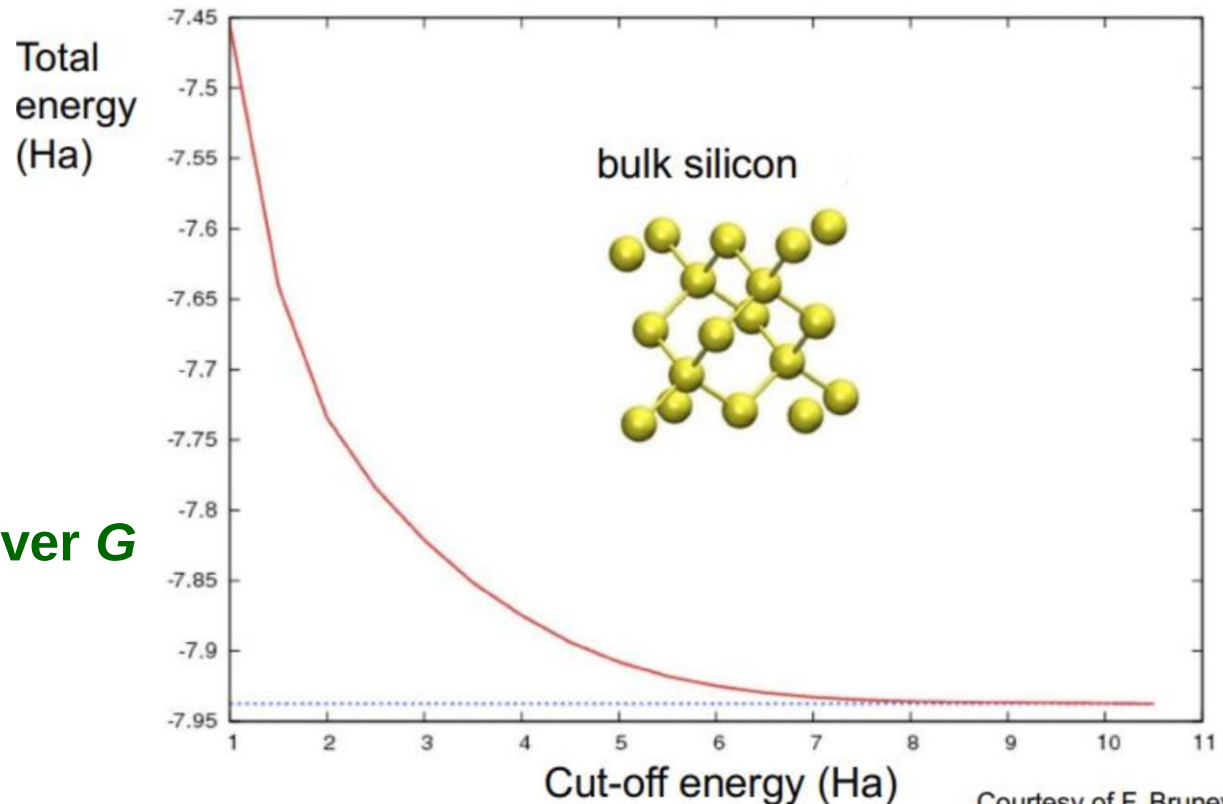
$$\psi_{jk}(r) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{G}} c_{j\mathbf{k}+\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})r}$$

Kinetic energy of a plane wave: $-\frac{\nabla^2}{2} (e^{i(\mathbf{k}+\mathbf{G})r}) = \frac{(\mathbf{k} + \mathbf{G})^2}{2} e^{i(\mathbf{k}+\mathbf{G})r}$

Larger gradients of wavefunction require larger $|\mathbf{k} + \mathbf{G}| \rightarrow$ larger plane wave basis set

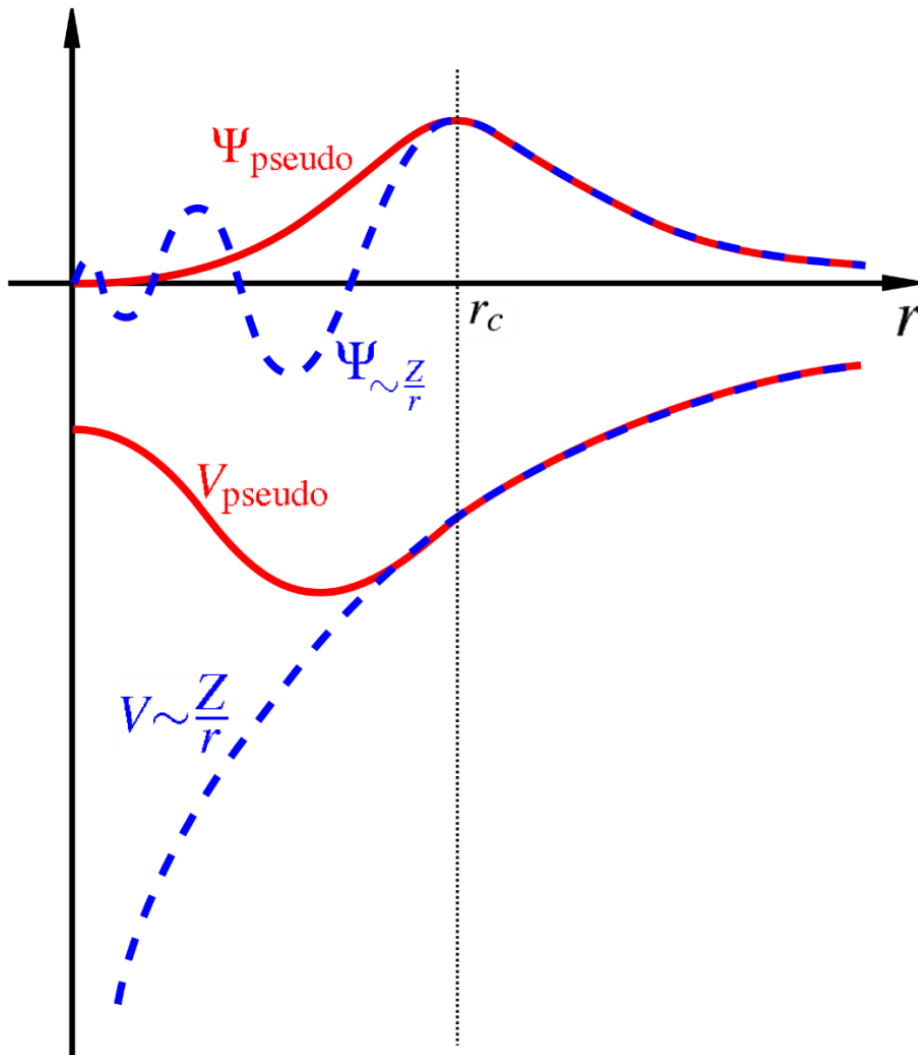
Need to limit the sum over $\mathbf{G}$

$$\rightarrow \frac{(\mathbf{k}+\mathbf{G})^2}{2} < E_{cut}$$



Courtesy of F. Bruneval

Pseudopotentials

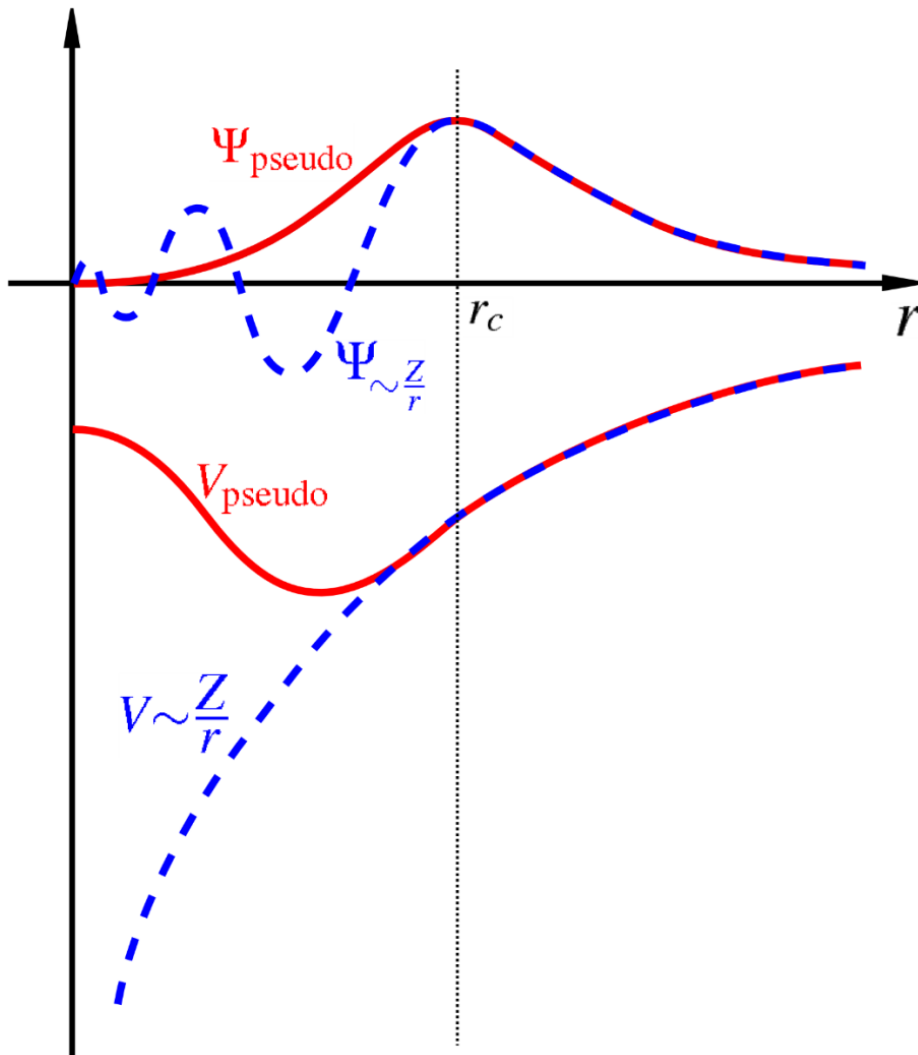


General idea: replace localized core electrons by such a potential V_{pseudo} that:

- 1) The solutions of Schrödinger equation with V_{pseudo} for an atom coincide with exact wavefunction for the same atom at $r > r_c$.
- 2) The electronic energies for pseudopotential and exact potential should be the same.
- 3) The pseudo-wavefunctions are smooth for all r .

Is this possible? Yes, but for each n/l the potential is different – OK, can be done.

Pseudopotentials

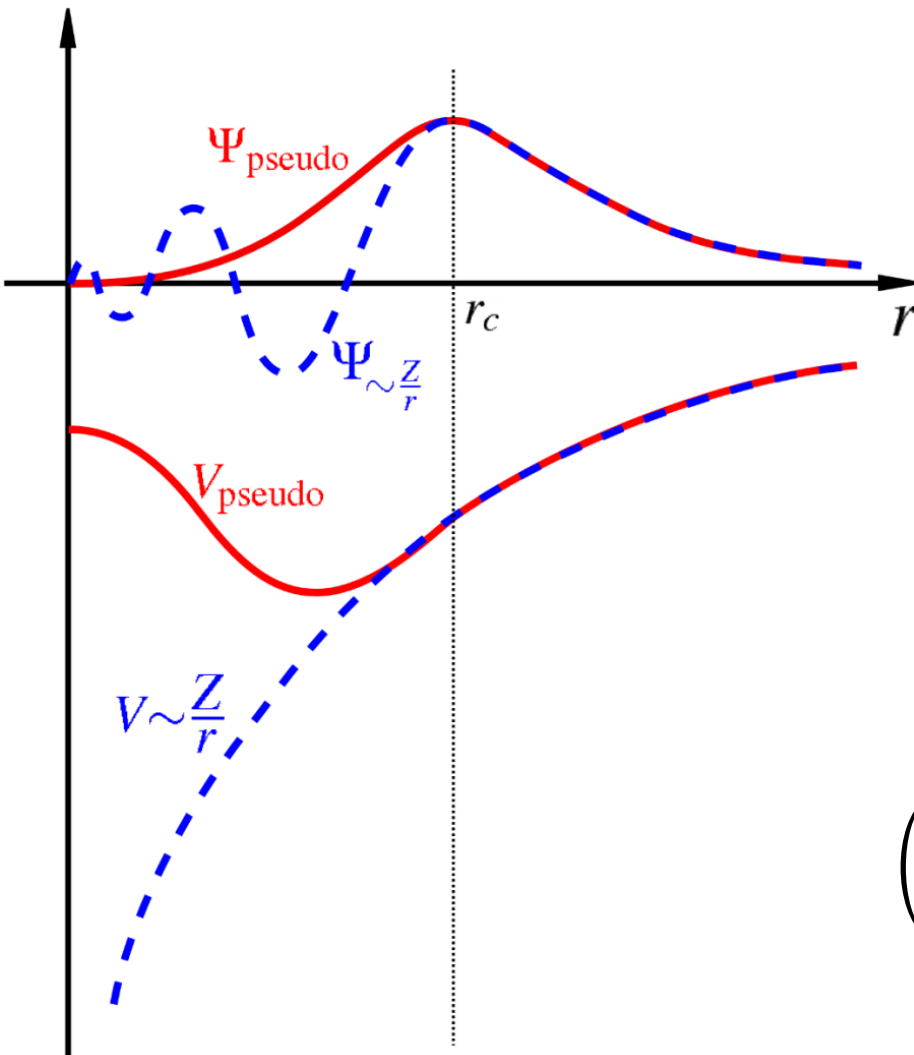


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- 1) The solutions of Schrödinger equation with V_{pseudo} for an atom coincide with exact wavefunction for the same atom at $r > r_c$.
- 2) The electronic energies for pseudopotential and exact potential should be the same.
- 3) The pseudo-wavefunctions are smooth for all r .

Some conditions can be relaxed to increase smoothness → from norm-conserving to ultra-soft pseudopotentials

Pseudopotentials



$$\phi_{nl}^{pseudo}(r) = \begin{cases} \sum_{k=1}^{N_b} c_{nlk} r j_l(q_k r), & r < r_c \\ \phi_{nl}(r), & r \geq r_c \end{cases}$$

The parameters c_{nlk} , q_k are found by maximizing smoothness of pseudo-wavefunction

The pseudopotential is then found by inverting Schrödinger equation:

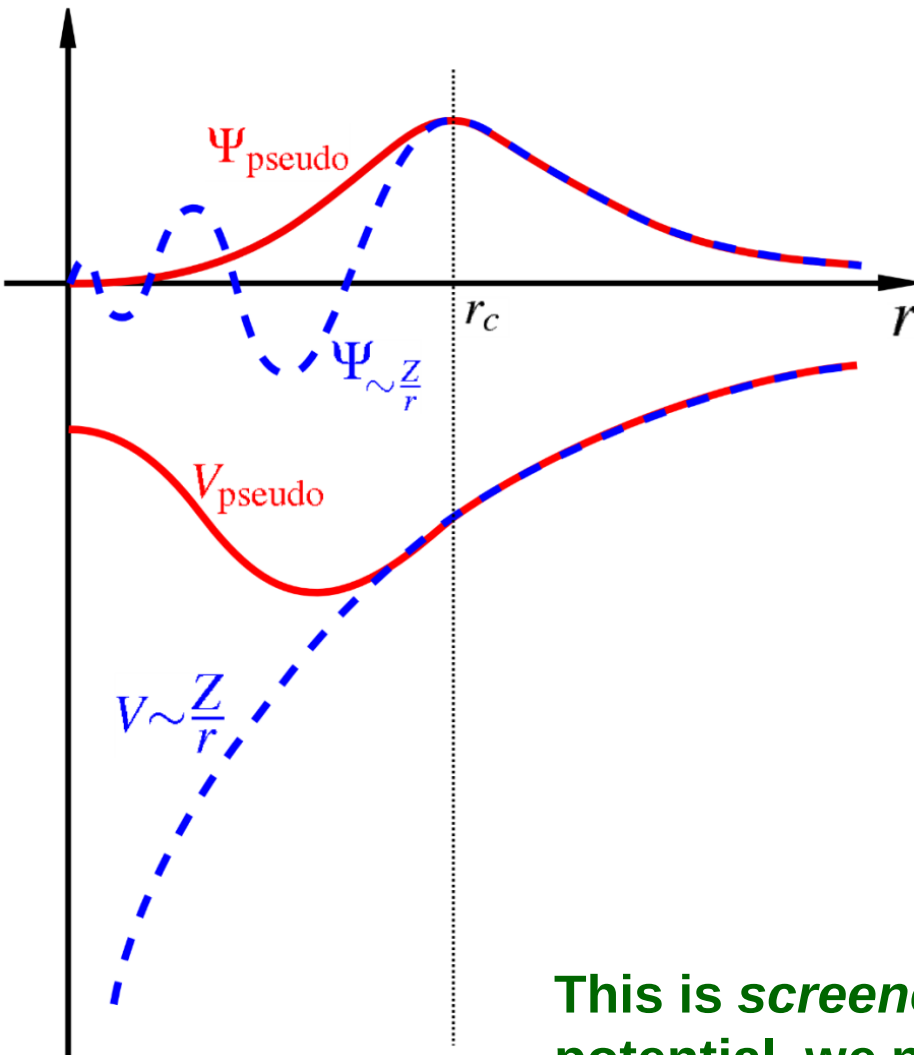
$$\left(-\frac{\nabla^2}{2} + V_{nl}^{pseudo} \right) \phi_{nl}^{pseudo} = \epsilon_{nl} \phi_{nl}^{pseudo}$$

$$V_{nl}^{pseudo} = \epsilon_{nl} + \frac{\nabla^2 \phi_{nl}^{pseudo}}{2\phi_{nl}^{pseudo}}$$

from all-electron calculation

has no nodes

Pseudopotentials



$$\phi_{nl}^{pseudo}(r) = \begin{cases} \sum_{k=1}^{N_b} c_{nlk} r j_l(q_k r), & r < r_c \\ \phi_{nl}(r), & r \geq r_c \end{cases}$$

The parameters c_{nlk} , q_k are found by maximizing smoothness of pseudo-wavefunction

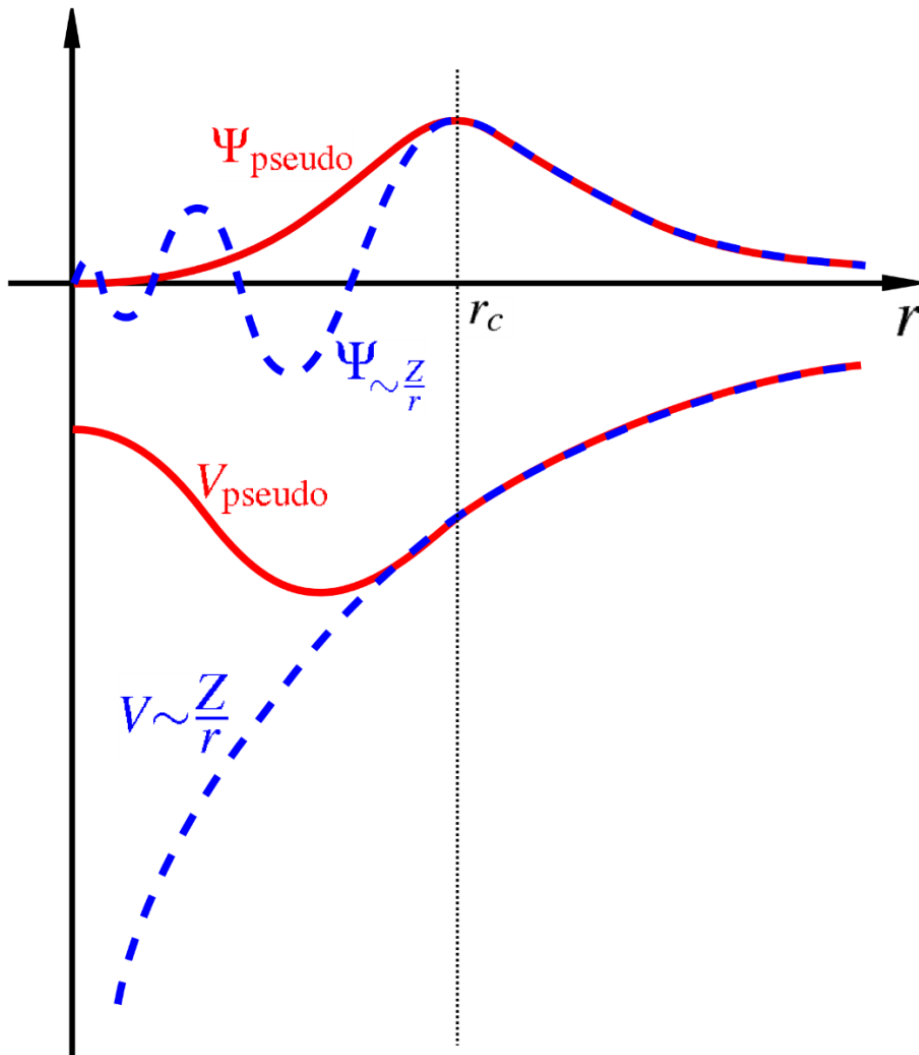
The pseudopotential is then found by inverting Schrödinger equation:

$$V_{nl}^{pseudo} = \varepsilon_{nl} + \frac{\nabla^2 \phi_{nl}^{pseudo}}{2\phi_{nl}^{pseudo}}$$

This is *screened* potential; to obtain ionic potential, we need to un-screen it:

$$V_{nl}^{ion} = V_{nl}^{pseudo} - V_H[\{\phi(r)\}_v] - V_{XC}[\{\phi(r)\}_v]$$

Pseudopotentials



Pseudopotentials allow us to reduce the cost of calculations (lower energy cutoff)

However, this is an *approximation*

As a rule, do not use pseudopotentials derived with one exchange-correlation functional, with another XC functional

At high pressures, the core regions may overlap, and pseudopotential approximation breaks

Good pseudopotentials for heavy atoms were developed very recently

Projector-augmented waves

Idea by P. Blöchl Phys. Rev. B 50, 17953 (1994):

The true wavefunction and a well-behaving pseudo-wavefunction are linked by a linear transformation

The diagram illustrates the linear transformation between the true wavefunction and the auxiliary wavefunction. The equation $\Psi = \hat{T}\tilde{\Psi}$ is shown, with Ψ and $\tilde{\Psi}$ circled in red. A red arrow points from the circled Ψ to the text 'True wavefunction (all electrons)', and another red arrow points from the circled $\tilde{\Psi}$ to the text 'Auxiliary wavefunction (soft pseudofunction)'. Red text annotations describe the behavior of each wavefunction near the nucleus.

$$\Psi = \hat{T}\tilde{\Psi}$$

Strong oscillations near the nucleus

No oscillations near the nucleus

True wavefunction
(all electrons)

Auxiliary wavefunction
(soft pseudofunction)

Projector-augmented waves

True wavefunction Ψ

Well-behaving pseudo-wavefunction $\tilde{\Psi}$

Linked by a linear transformation $\Psi = \hat{T}\tilde{\Psi}$

Physical quantities can be calculated as $\langle \Psi | \hat{A} | \Psi \rangle$

original operator is
replaced by pseudo
operator

Computed in the pseudo representation $\langle \tilde{\Psi} | \tilde{A} | \tilde{\Psi} \rangle$ with $\tilde{A} = \hat{T}^\dagger \hat{A} \hat{T}$

For total energy can do the same, then apply variational principle
solve Schrödinger equation:

$$\frac{\delta \langle \tilde{\Psi} | \hat{T}^\dagger \hat{H} \hat{T} | \tilde{\Psi} \rangle}{\delta \tilde{\Psi}^*} = 0$$



Equivalent to Kohn-Sham equations for pseudo-wavefunctions
Search for ground state done in the pseudo space
The oscillating part is expanded in a Gaussian basis set

Projector-augmented waves

$$\psi_{\text{AE}} = \psi_{\text{PP}} + \sum_i (\phi_i - \tilde{\phi}_i) \langle o_i | \psi_{\text{PP}} \rangle$$

solutions for the atomic potential solutions for the pseudoatom projector functions

$$o_i = \sum_j f \tilde{\phi}_j \alpha_{ij}$$

a cutoff function

$$\langle o_i | \tilde{\phi}_j \rangle = \sum_k \alpha_{ik} \langle \tilde{\phi}_k | f | \tilde{\phi}_j \rangle = \delta_{ij} \rightarrow \alpha_{ik}$$



projector functions o_i

Projector-augmented waves

Essentially all-electron, structure and energetics on par with all-electron full-potential methods

It is still not fully all-electron, because the localized orbitals are frozen to atomic orbitals → problems when core polarization is important

Concluding remarks

- 1) Periodic models can be efficiently used to study concentration/coverage dependence, including infinitely dilute limit (low-dimensional systems, defects, etc.)
- 2) A lot of useful and experimentally testable information on material's properties can be obtained from the analysis of its electronic structure (band structure, DOS, APDOS, etc.)
- 3) A lot of development (in both computational methods and code efficiency) is still necessary to go beyond standard DFT for periodic systems, and to approach accuracy that can be achieved nowadays for molecules

Recommended literature

Neil W. Ashcroft and N. David Mermin, “*Solid state physics*”

Axel Groß, “*Theoretical surface science: A microscopic perspective*”

Roald Hoffmann (1981 Nobel Prize in Chemistry (shared with Kenichi Fukui)):

- 1) “*How Chemistry and Physics Meet in the Solid State*”, Angew. Chem. Int. Ed. Engl. 26, 846-878 (1987)
- 2) “*A chemical and theoretical way to look at bonding on surfaces*”, Reviews of modern physics, 60, 601-628 (1988)